

# Human vs. AI in Symbolic Music: An Online Listening Study on Bach-Style Chorales

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## Introduction & Method

We conducted a music perception study to evaluate human vs. AI discrimination in Bach-style chorales. Participants ( $N = 295$ ) listened to **20 isolated musical phrases** and categorized each as **AI-** or **human-composed**. AI snippets were generated using *NotaGen* [1], a state-of-the-art symbolic music generation model. All musical snippets (both human and AI) were manually curated. Participation was open to the public with **no prior music qualification required**. Prior to and following the experiment, we collected metadata such as age, musical background, and self-reported confidence levels in order to better contextualize participant performance.

## Experiment Design

**Pre-Survey** → **20 Listening Trials** → **Post-Survey**  
(Order randomized)

### Stimuli

- ▶ **Balanced Set:** 10 Bach chorale phrases vs. 10 *NotaGen* phrases
- ▶ **Audio Uniformity:** All excerpts rendered with an identical MIDI harpsichord soundfont
- ▶ **Selection:** All snippets were manually curated

### Measured Parameters

- ▶ **Demographics:** Age, weekly AI usage (Yes/No).
- ▶ **Musical Background:** Plays instrument (Yes/No), sings in choir (Yes/No), self-reported experience (4-point Likert scale).
- ▶ **Confidence:** Pre- and post-experiment self-ratings (4-point Likert scale).

## Overview and Results

### Participant Overview

- ▶ **Sample Size:**  $N = 295$  participants (primarily German, aged 15–86).
- ▶ **Musical Background:**  $\approx 90\%$  play an instrument;  $> 50\%$  sing in a choir.
- ▶ **AI Usage:**  $\approx 50\%$  use AI multiple times per week.

### Participant Feedback<sup>1</sup>

- ▶ "frighteningly difficult 😬"
- ▶ "It was enormously difficult to hear a difference. I was constantly guessing. If I guessed something correctly, it merely was a gut instinct. [...]"
- ▶ "It was a very interesting experience because I realized, how damn hard it is to me distinguish AI compositions from human compositions."
- ▶ "The snippets are relatively short. Longer snippets could create a higher hit rate, I guess..."
- ▶ "It's really exciting because it's really difficult to decide whether it was AI or human after hearing it only once. Sometimes I would have needed it a second time."

<sup>1</sup>Quotes were translated from German to English

## Age Distribution

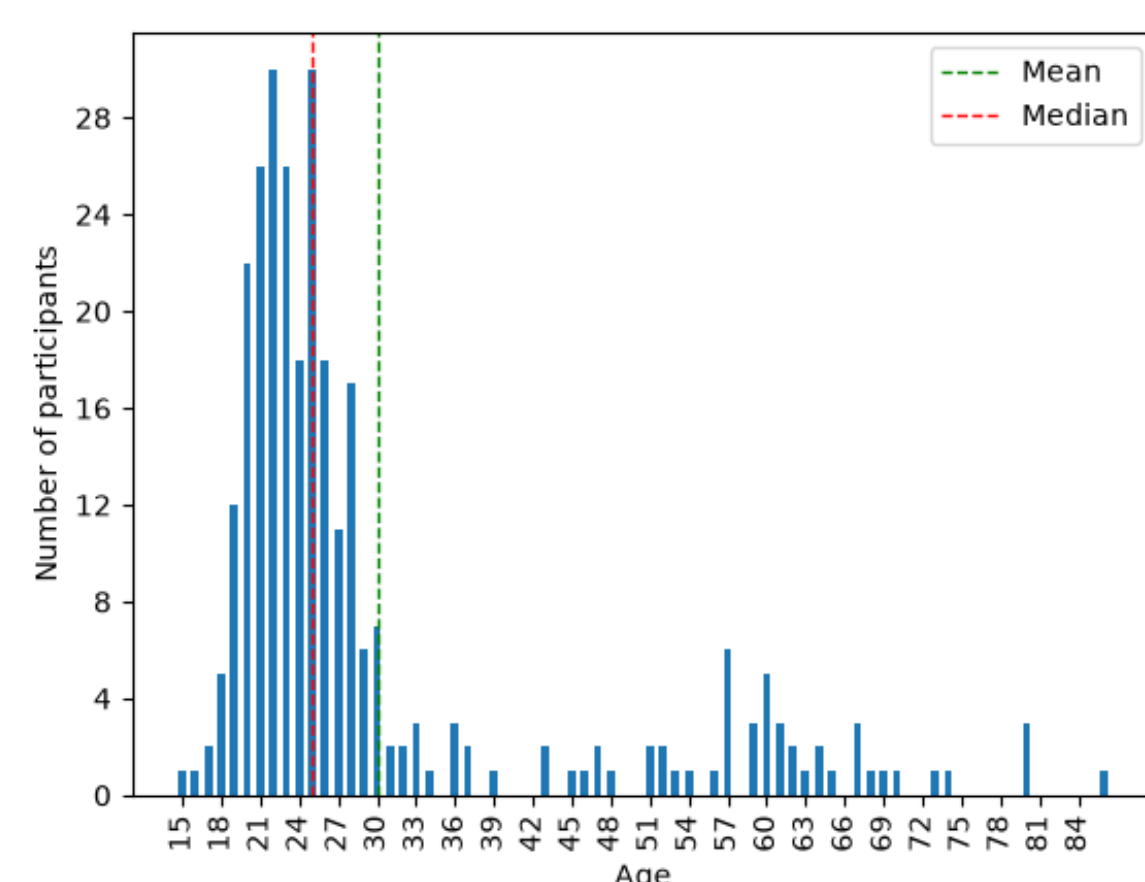


Figure 1: Participant age distribution ( $N = 295$ )

## Performance by Choir Experience

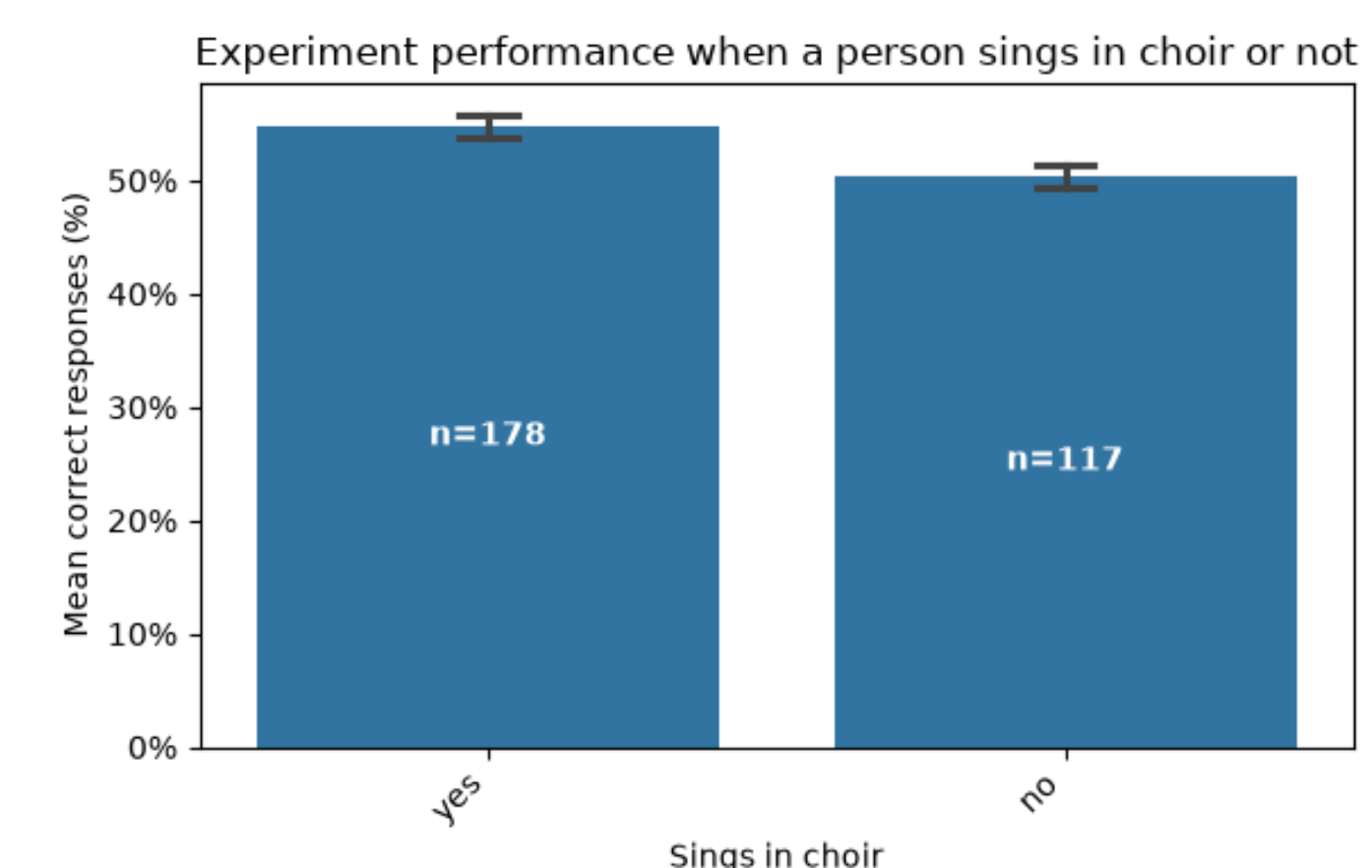


Figure 3: Accuracy comparison by participants' choir experience

## Performance by Musical Experience

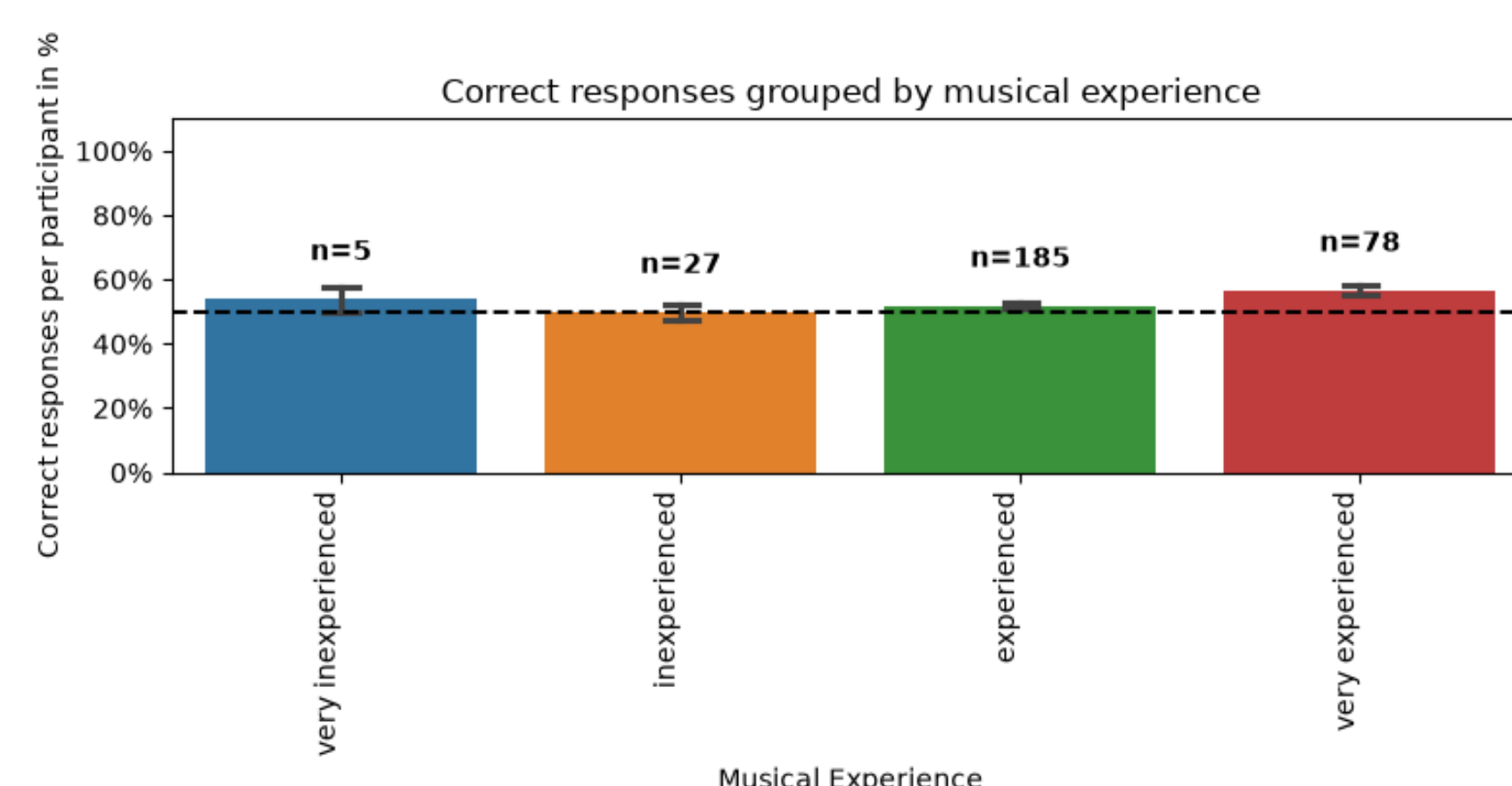


Figure 2: Discrimination accuracy across participants' prior musical experience

## Confidence Shifts

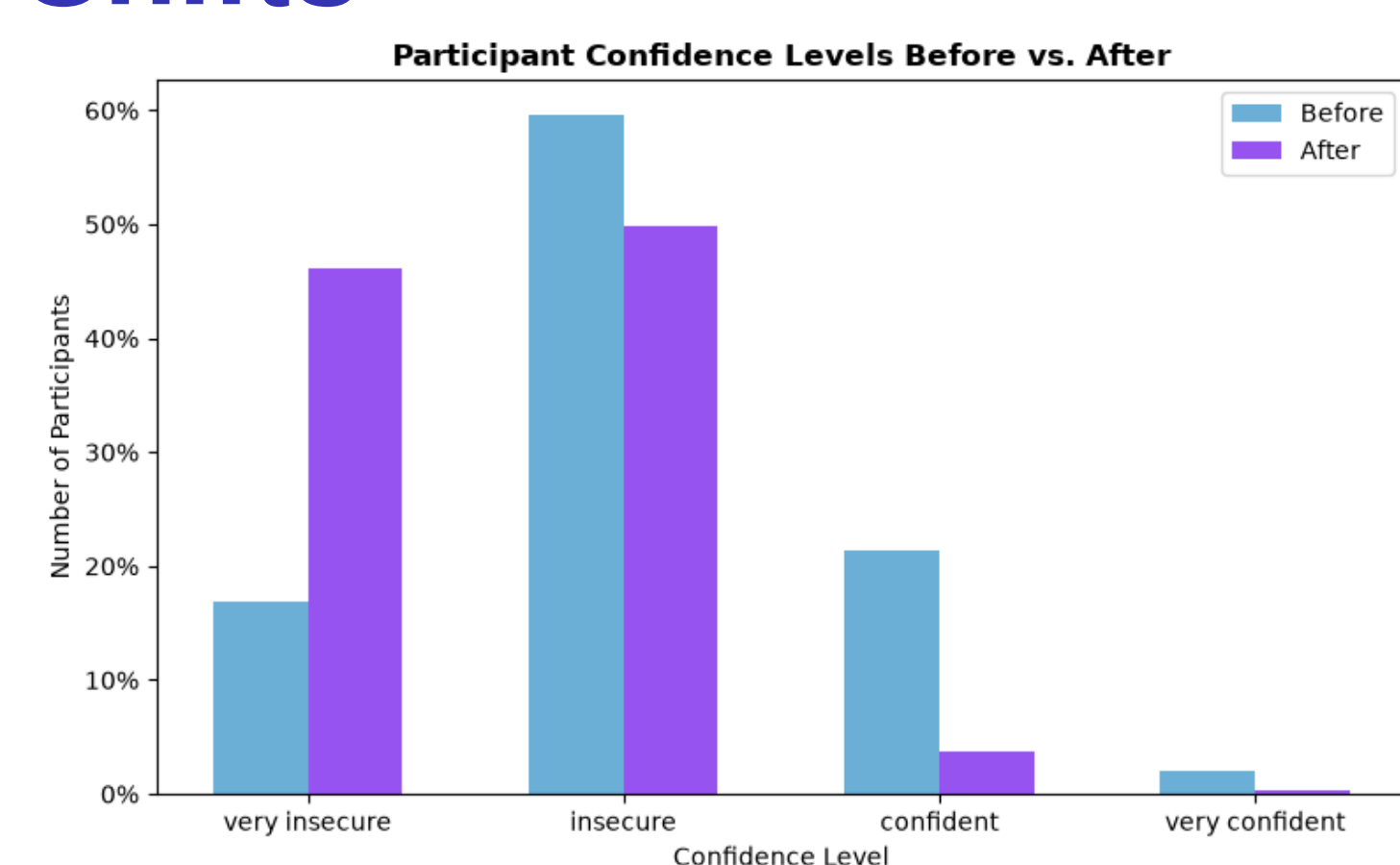


Figure 4: Self-reported confidence pre- vs. post-trial.

## Overview: Performance Per Participant

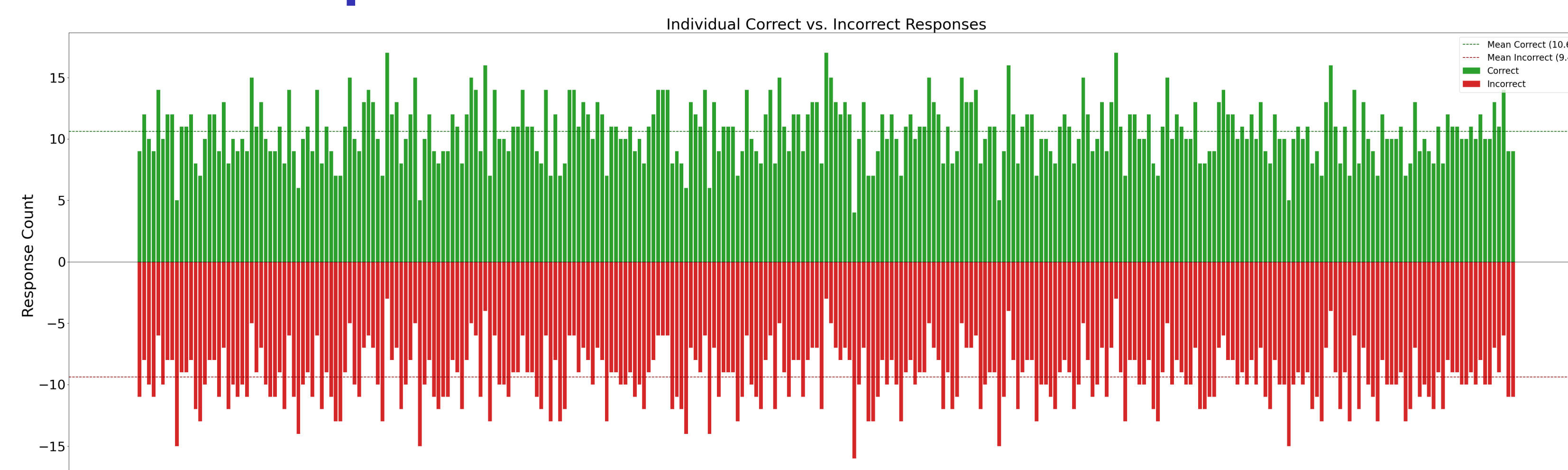


Figure 5: Individual participant accuracy scores across all trials.

## References

- [1] Yashan Wang et al. *NotaGen: Advancing Musicality in Symbolic Music Generation with Large Language Model Training Paradigms*. Mar. 2025. DOI: 10.48550/arXiv.2502.18008. arXiv: 2502.18008 [cs]. URL: <http://arxiv.org/abs/2502.18008> (visited on 01/21/2026).

## Try the Experiment



Can you distinguish AI from human-composed?

← Scan to test your skills in our study

[jasonvla.github.io/ai-chorales](https://jasonvla.github.io/ai-chorales) website